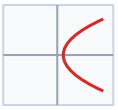
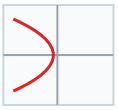
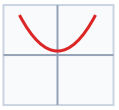
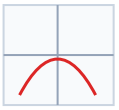
WORKSHEET: CHAPTER - POLYNOMIALS

Class: 10th | Subject: Mathematics

Q 1. Which of the following graphs could be for the simple polynomial x^2?

(a)  (b)  (c)  (d) 

Q 2. How many zero(s) does the polynomial $293x^2 - 293x$ have? [CBSE Question Bank]

(a) 0 (b) 1 (c) 2 (d) 3

Q 3. The number of polynomials having zeros 1 and −2 is:

(a) 1 (b) 2 (c) 3 (d) more than 3

Q 4. A quadratic polynomial, whose zeros are − 3 and 4, is:

(a) $x^2 - x + 12$ (b) $x^2 + x + 12$ (c) $^{x^2}/_2 - {}^{x}/_2 - 6$ (d) $2x^2 + 2x - 24$

Q 5. The zeros of the quadratic polynomial $x^2 + kx + k$, $k \neq 0$

(a) cannot be positive (b) cannot be negative

(c) are always equal (d) are always unequal

Q 6. If the graph of a polynomial intersects the x-axis at exactly two points, then it:

(a) cannot be a linear or a cubic polynomial (b) can be a quadratic polynomial only

(c) can be a cubic or a quadratic polynomial (d) can be a linear or a quadratic polynomial

Q 7. p and q are the zeros of the polynomial $4y^2 - 4y + 1$. What is the value of $^1/_p + {}^1/_q + pq$? [CBSE Question Bank]

(a) $^{-15}/_4$ (b) $^{-3}/_4$ (c) $^5/_4$ (d) $^{17}/_4$

Q 8. How many zero(s) does $(x - 2)(x + 3)$ have?

(a) zero (b) one (c) two (d) three

Q 9. A part of the graph of a polynomial $h(x)$ is given alongside. On dividing $h(x)$ by which of the following will the remainder be zero? [CBSE Question Bank]

(i) $(x - 2)$ (ii) $(x + 2)$

(iii) $(x - 4)$ (iv) $(x + 4)$



(a) only (ii)

(b) only (i) and (iii)

(c) only (ii) and (iv)

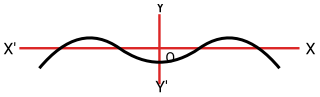
(d) cannot be determined without knowing the polynomial $h(x)$

The following question is Assertion and Reason based question. Select the correct answer from options (a), (b), (c) and (d).

Q 10.

Assertion (A): The graph $y = f(x)$ is shown in figure, for the polynomial $f(x)$. The number of zeros of $f(x)$ is 4.

Reason (R): The number of zero of the polynomial $f(x)$ is the number of point at which $f(x)$ cuts or touches the axes.



Q 11. If p and q are the zeros of the polynomial $f(x) = 2x^2 - 7x + 3$, find the value of $p^2 + q^2$.

Q 12. If α and β are the zeros of the quadratic polynomial $f(x) = 3x^2 - 5x - 2$, then evaluate $\alpha^2/\beta + \beta^2/\alpha$.

Q 13. If m and n are the zeros of the polynomial $3x^2 + 11x - 4$, find the value of $m/n + n/m$.

Q 14. If α and β are the zeros of the quadratic polynomial $f(x) = x^2 - px + q$, Prove that $\alpha^2/\beta^2 + \beta^2/\alpha^2 = p^4/q^2 - 4p^2/q + 2$.

Q 15. Find the zeros of the polynomial $x^2 + 1/6x - 2$, and verify the relation between the co-efficients and zeros of the polynomial.

Q 16. If α and β are the zeros of the quadratic polynomial $f(t) = t^2 - p(t + 1) - c$, Prove that $(\alpha + 1)(\beta + 1) = 1 - c$.

Q 17. Find the zeros of the quadratic polynomial $x^2 - (2a + b)x + 2ab$ and verify the relationship between the zeros and the coefficients.

Q 18. If the sum of squares of the zeros of the quadratic polynomial $x^2 - 8x + k$ is 40, find k.

Q 19. If α and β are zeros of polynomial $f(x) = 2x^2 + 11x + 5$, then find: (i) $\alpha^4 + \beta^4$ (ii) $1/\alpha + 1/\beta - 2\alpha\beta$

Q 20. For a linear polynomial $kx + c$, $k \neq 0$, the graph of $y = kx + c$ is a straight line which intersects the x-axis at exactly one point, namely, $(-c/k, 0)$. Therefore, the linear polynomial $kx + c$, $k \neq 0$, has exactly one zero, namely, the x-coordinate of the point where the graph of $y = kx + c$ intersects the x-axis.

Answer the following questions:

(i) If a linear polynomial is $2x + 3$, then what is the zero of $2x + 3$?

(ii) The graph of $y = p(x)$ is given in figure below for some polynomial $p(x)$. What is the number of zero/zeros of $p(x)$?

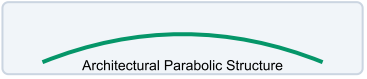


(iii) (a) If α and β are the zeros of the quadratic polynomial $x^2 - 5x + k$ such that $\alpha - \beta = 1$, then find the value of k.

(b) If α and β are the zeros of the quadratic polynomial $p(x) = 4x^2 + 5x + 1$, then find the product of zeros.

Q 21. The below pictures are few natural examples of parabolic shape which is represented by a quadratic polynomial. A parabolic arch is a curve in the shape of a parabola. This shape distributes load evenly across the arch, making it ideal for use in bridges and a variety of other architectural structures.





Based on the above information, answer the following questions:

(i) If α and $1/\alpha$ are the zeros of the quadratic polynomial $2x^2 - x + 8k$, then write the value of k.

(ii) Find the sum of zeros of the polynomial $p(x) = 2x^2 - x + 2$.

(iii) (a) If the sum of the zeros is $-p$ and product of the roots is $-1/p$, then find the quadratic polynomial.

(b) Write the zeros of the polynomial $p(x) = 4x^2 - 4x + 1$.